Hsin-Ling (Justin) Hsu

Google Scholar: Hsin-Ling Hsu
GitHub: JustinHsu1019
LinkedIn: justinhsu101999
justin-code.com
X (Twitter): JustinHsu99
justin.hsu.1019@gmail.com

Education

National Chengchi University (NCCU) Sep. 2023 – Expected Graduation: Jun. 2027
B.S. Double Major in MIS and Computer Science *Taipei, Taiwan*

- GPA: 4.11 / 4.30 (3.94 / 4.00) | Honors: Beta Gamma Sigma Honor Society (Top 10% of department).
- Transferred from NCCU Department of Mathematical Sciences after freshman year.

Research Interests

Trustworthy AI for language models deployed in complex, high-stakes real-world applications such as healthcare and autonomous systems, spanning interpretability, verifiable repair, and reliability.

Publications

Total citations: 60; h-index: 2; i10-index: 2 (Google Scholar, July 2026)

[1] **Hsin-Ling Hsu**, et al., *"MedAction: Towards Active Multi-turn Clinical Diagnostic LLMs." Third Conference on Language Modeling (**COLM** 2026)*, San Francisco, USA, 2026. *(paper)*
Notes: Top conference, ∼**29%** acceptance rate.

[2] **Hsin-Ling Hsu***, Cong-Tinh Dao*, et al., *"MedPlan: A Two-Stage RAG-Based System for Personalized Medical Plan Generation." Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL** 2025): Industry Track*, Vienna, Austria, 2025. *(paper)*
Notes: A* top conference, ∼**25.6%** acceptance rate. 30+ citations.

[3] Chia-Hsuan Hsu*, Jun-En Ding*, **Hsin-Ling Hsu**, et al., *"RGPO: Ranking-Guided Preference Optimization for Reliable Clinical Reasoning." IEEE Transactions on Artificial Intelligence (**IEEE TAI**, Q1 Journal)*, 2026. *(paper)*

[4] **Hsin-Ling Hsu**, et al., *"KAP: MLLM-assisted OCR Text Enhancement for Hybrid Retrieval in Chinese Non-Narrative Documents." Proceedings of the FinTech in AI CUP Special Session, 18th **NTCIR** Conference*, Tokyo, Japan, 2025. *(paper)*
Notes: Oral Presentation. Travel grant from ROC Ministry of Education ($3,600 USD). National Excellence Prize at AI CUP 2024.

Patents

- *Electronic Device for Generating Personalized Assessment Content and Treatment Plan*, Taiwan (R.O.C.) Utility Model Patent No. M682510, granted May 2026. First Inventor.

Research Experience

Research Collaborator Feb. 2026 – Present
BioMed-AI Lab, ECE/EECS, University of Michigan (Advisor: Prof. Liyue Shen) *Remote*

- Research on AI for healthcare, focusing on multi-turn clinical diagnostic reasoning with LLMs, knowledge graphs, and chain-of-reasoning-action consistency. [1][8]

Research Intern Nov. 2025 – Present
FLAIR Lab, CSE, Texas A&M University (Advisor: Prof. Kuan-Hao Huang) *Remote*

- Research on vision-language alignment, interpretability, and hallucination in VLMs, focusing on cross-modal representations via sparse autoencoders.

Research Assistant Dec. 2024 – Present
Far Eastern Memorial Hospital (Advisor: Dr. Fang-Ming Hung) *New Taipei, Taiwan*

- Research on explainable clinical decision support systems, integrating VLMs/LLMs, reinforcement learning, and knowledge graphs for disease/ICD prediction and medical plan generation. [1][2][3][8]

Industry Experience

Engineer (Part-Time) May 2024 – July 2026
RLG Lab, IIS, Academia Sinica (Advisor: Prof. Ti-Rong Wu) *Taipei, Taiwan*

- Reinforcement learning infrastructure: game environments and AI solution tree visualization.

AI Intern Sep. 2024 – Jun. 2025
GoFreight (The world's largest cloud-based freight forwarding software) *Taipei, Taiwan*

- Leveraged MLLM parallelization for logistics OCR/NLP, reducing processing time by **~67%** (45s to 15s).
- Built LLM-based web parsers with up to **97%** accuracy.

AI Engineer (Part-Time) Jul. 2023 – Sep. 2024
ChainSea Information Group *Taipei, Taiwan*

- Core R&D on LLM/Whisper projects (transcription, inference acceleration, LoRA tuning).
- Built RAG pipelines and an LLM Agent for addiction counseling.

Selected Competitions & Honors

22nd National Innovation Award 2025
Project: MedPlan (ACL 2025). Co-participating with Far Eastern Memorial Hospital. *Taiwan*

- First Inventor of patent. MOU signed with clinics/system vendors; deployed in clinical trials at hospital.

2nd Place, HOTAI MaaS Hackathon, [2/233 teams; ~0.8%] 2024
*AI Travel Itinerary Health Check. Won **NT$250,000** prize. Tech: React.js, Hybrid RAG, Cross-Encoder.* *Taiwan*

3rd Place, LINE FRESH Campus Competition, [3/165 teams; ~1.8%] 2024
AI dementia care platform. Built backend, health tracking, and multilingual chatbot with LLMs. *Taiwan*

CPE Collegiate Programming Examination — Professional Level (A), Top 4.6% 2025
Ranked 113/2481. ICPC-style algorithmic problem solving. *Taiwan*

Project Experience

MedPlan: RAG-Based Personalized Medical Plan Generation Dec. 2024 – Present
Far Eastern Memorial Hospital *New Taipei, Taiwan*

- Designed **MedPlan** [2], a SOAP-aligned two-stage architecture that first generates a clinical assessment, then formulates a patient-specific treatment plan via retrieval-augmented generation, mirroring real clinician reasoning workflows.
- Published at ACL 2025 Industry Track ($\sim$25.6% acceptance), accumulating 30+ citations; awarded the 22nd National Innovation Award and a Taiwan utility model patent (No. M682510, First Inventor).
- Deployed the platform clinically at Far Eastern Memorial Hospital, validated on real patient records by attending physicians; achieved 90+% ICD-10 prediction accuracy, with ongoing specialty-stratified refinement.
- Signed MOUs with clinics and HIS vendors to advance adoption; partnerships currently progressing toward broader deployment.

MedAction: Active Multi-turn Clinical Diagnostic LLMs Feb. 2026 – June 2026
BioMed-AI Lab, University of Michigan & Far Eastern Memorial Hospital *Remote*

- Built a tree-structured distillation pipeline and two KG-grounded metrics (DTC, RAC) to synthesize and filter high-quality multi-turn diagnostic trajectories for **MedAction** [1], yielding MedAction-32K (32,681 trajectories, 2,896 PMC cases); fine-tuned 8B model achieves SOTA among open-source models on MedRBench and a curated hard benchmark. Accepted to COLM 2026 ($\sim$29% acceptance rate).
- Extended the work into **MedAction-RL** [7], a post-training and scaling recipe (environment construction + SFT curation + RL) for active diagnostic agents that order tests, interpret results, and decide when to stop; achieves SOTA open-source diagnostic accuracy with improved test-ordering precision and cost.
- Identified key scaling findings: filtered turn-level SFT data cold-starts active diagnosis better than raw teacher trajectories, and model/environment scaling trends diverge across general vs. medical backbones.

WARP & FORGE: Verifiable Repair of NLP Transformers Sep. 2025 – July 2026
Advisor: Prof. Fang Yu, National Chengchi University (Undergraduate Thesis) *Taipei, Taiwan*

- Proposed **WARP** [5], the first provable repair framework for Transformer-based NLP models beyond the final layer, formulating repair as a convex QP with per-sample margin, preservation, and certified robustness-radius guarantees.
- Extended the framework into **FORGE** [6], generalizing verifiable repair to generative LLM behaviors (bias, toxicity) by decoupling defect localization, weight editing, and behavioral verification, and supporting WARP-style constrained optimization and null-space projection as interchangeable backends.
- Demonstrated FORGE outperforms gradient-based fine-tuning in bias/toxicity reduction with minimal perplexity degradation, and remains stable in low-resource settings where standard fine-tuning fails to converge.
- Awarded National Science and Technology Council (NSTC) Undergraduate Research Grant for this project.

Under Review

[5] **Hsin-Ling Hsu**, Min-Yu Chen, Nai-Chia Chen, Yan-Ru Chen, Yi-Ling Chang, Fang Yu, *"WARP: Guaranteed Inner-Layer Repair of NLP Transformers." Submitted to AAAI 2027. (paper)*

[6] **Hsin-Ling Hsu**, Min-Yu Chen, Nai-Chia Chen, Yan-Ru Chen, Yi-Ling Chang, Fang Yu, *"FORGE: Verification-Gated Behavioral Repair for Generative Language Models." Submitted to ICSE 2027*.

[7] **Hsin-Ling Hsu***, Bohao Wang*, Zizheng Wang*, Pengfei Hu, Jun-En Ding, Xiaoyu Qiu, Donghua Zhang, Yichen Guo, David Restrepo, Luis Filipe Nakayama, Liyue Shen, Chenwei Wu, *"MedAction-RL: Post-Training and Scaling of Active Medical Diagnosis Agents." Submitted to EMNLP 2026 Industry Track*.

Academic Service & Other Activities

- **Reviewer / Program Committee:** IEEE TCE Journal, KDD 2026 AI4Sci Track, Texas NLP Symposium 2026, WMW 2026, NeurIPS 2025 Workshop on GenAI for Health, NeurIPS 2025 Workshop on Efficient Reasoning.
- **Talks:** PyCon TW (academic research sharing); Google Developer Group at NCCU (Generative AI, Databases).
- **Volunteer:** Soobi rural children's programming camp instructor (Meowbit game design).
- **Columnist:** *Customer Insights* magazine (two columns on generative AI applications).
- **Open Source:** Core contributor to an LLM project with 300+ GitHub Stars and 30+ forks.
- **Security:** Two ZeroDay vulnerability disclosures.
- **Industry Projects:** Taipower smart AI chatbot ("Dianbao") training data optimization (during tenure at ChainSea).

Skills

Python, C/C++, PyTorch, Unsloth, veRL, Transformers, Flask, Langfuse, SQL, Linux, GCP, Docker, Git, HTML/CSS/JS